DAY-2

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

- **No display on the monitor:** The computer turns on, but the monitor shows a blank screen and nothing appears.
- **Faulty RAM:** The computer keeps restarting or makes beep sounds and does not start properly.
- **Hard disk/SSD failure:** The computer becomes very slow or Windows does not load.
- **Power supply (SMPS) problem**: The computer does not turn on when the power button is pressed.
- **CPU overheating:** The computer becomes very hot, freezes, or restarts automatically while in use.

2.Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

- **Check the power socket** – make sure the wall socket is working by plugging in another device.
- **Check the charger** – check if the charger cable or adapter is damaged or loose.
- **Check the charging port** – look for damage in the laptop's charging port.
- **Check the charging light** – see if the charging LED turns on when the charger is connected.

- **Test with another charger** – If available, try a compatible charger to check if the original charger is faulty.
- **Check the battery** – If the battery is removable, I would remove and reinsert it properly.

3.Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

1. First set the multimeter to DC Voltage (20V) mode.
2. Then connect the charger to a power socket.
3. After connection to socket placed the red probe on the USB positive (+) pin and the black probe on the negative (−) pin.
4. Then the multimeter displayed a voltage of 5.08 V.
5. compared the reading with the expected USB range of 4.75 V to 5.25 V.
6. Since 5.08 V is within the expected range, the charger is working normally.

Observation

Measured Voltage	Expected Range	Result
5.08 V	4.75 V – 5.25 V	Within the expected range (Normal)

**4.A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires..**

- First turned off the computer and disconnect the power supply safely.
- Then prepared the multimeter to measure DC voltage.
- Check the 12V output of the power supply following the manufacturer's safety instructions.

- Then measured voltage was 12.08 V.
- Compared the reading with the expected value of about 12 V.
- 12.08 V is within the normal range, the power supply was working properly.

Expected Voltage	Measured Voltage	Result
12 V	12.08 V	Normal